The Adjudication Problem

Ten months of temperature excursion handling at Kestrel, written down so that somebody can tell us what to do about it.

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CLASSIFICATION	Confidential

This is not a specification.

It is the situation as it stands, assembled from our own records. We have deliberately not written down what we think should be built, what it should do, or what properties it should have. Reading the problem, deciding what it actually is, and deciding which parts of it are worth solving first: that is the substance of what we are asking from you. If you finish and your conclusion is that we have asked for the wrong thing, that is a legitimate answer and we would like to hear it.

SECTION 1

Kestrel, in one page

Kestrel Cold Chain Solutions moves temperature-controlled pharmaceutical product for 34 enterprise customers (manufacturers, distributors and two hospital networks) out of operating hubs in Gurugram, Amsterdam, Newark and São Paulo. Eighteen of those customers are Indian, nine European, five American, two Brazilian. Last month we moved 128,000 temperature-controlled consignments and ingested about ten million sensor readings a day.

We do not really sell trucks and cold rooms; there are cheaper people for that. What a customer buys from us is the evidence that lets their pharmacist release the product at the other end. The product belongs to them and so does the regulatory obligation. What we hold is the custody record: what the temperature was, who had the pallet, what the paperwork says, and whether all of that hangs together well enough to survive an inspection. That is the product. Our platform is called Sentinel and it is priced at USD 3.10 per consignment, with cold chain compliance bundled into that figure rather than sold separately.

Everything in this document concerns one part of that: what happens when the temperature goes where it should not.

SECTION 2

What happens when the temperature goes out of range

Every consignment travels with a logger and moves through cold rooms that have their own telemetry. When a reading falls outside the product’s labelled storage range, Sentinel raises an excursion alert. We raised 11,400 of them last month. Roughly 82 per cent came to nothing: a logger left on a dock, a gateway that dropped out and came back, a chiller cycling, a permitted deviation that was always within the product’s tolerance. The remaining two thousand or so are real, and each one ends in a decision that a named human being signs: release the product, release

it with a shortened expiry, quarantine it pending a laboratory test, rework or relabel it, send it back, or destroy it.

Median time from alert to signed disposition is 6.2 days. That is a different clock from the regulatory one. In two of our markets we have 72 hours from the alert to notify the regulator that a deviation occurred; the disposition itself follows later, whenever the evidence is in. Missing the first is a compliance failure in its own right, whatever the second turns out to be.

A destroy decision usually condemns the affected pallets rather than the whole consignment. Last year that came to USD 19.4M of product across roughly 2,700 destroy decisions, and internal audit believes about a third of it was salvageable and went because nobody could assemble the evidence in time. Here is one case, in full, because the aggregate numbers do not convey the shape of the problem.

EXC-107133, Tuesday 28 July

At 06:14 UTC a logger travelling with fourteen pallets of Glucodyne SoloJet (insulin, labelled 2–8 °C, consignment value USD 96,400) reported 11.2 °C. It stayed above 8 °C for fifteen minutes and then came back. The shipment was on the second leg of a Bhiwandi–Kundli move. Sentinel raised the alert and put it in the queue.

Priya Menon picked it up at 09:30 UTC, a little after three in the afternoon in Gurugram. She has reviewed cases here for nine years and there were 186 alerts in the queue ahead of it, most of which she knew from the shape of them would come to nothing. What she had to establish before she could sign anything was whether fifteen minutes at 11.2 °C mattered for this product. The label says 2–8 °C. The label is not the answer. The answer is in the product’s stability file, which says how long this presentation tolerates what temperature and whether exposure accumulates over the shelf life. She opened the stability repository, which is a SharePoint folder, and searched for Glucodyne. Four documents came back. Two were for the vial, not the pen. One was a scan of a scan and she could not read the table. The fourth was the right study and it permitted a single excursion of up to six hours below 15 °C.

That should have closed it. It did not, for four reasons.

The product reference. The alert carried 08901234500035, a GTIN. The consignment note said “Glucodyne 100U”. The customer’s ERP feed calls it SKU-INS-0119. The stability study is filed under the brand name with the strength spelled differently again. Priya knows these are the same thing. Nothing in Sentinel does.

The acceptance limit. What governs a release at Kundli is not the stability file. It is the limit in the customer’s own SOP: revision controlled, site specific, and in force on the date of the excursion rather than today. Docuvault returned three revisions of “Temperature Excursion Management” for this customer with overlapping effective dates. The site-specific limits live in Annexure 2, filed as a separate document. For two of the six of our sites this customer ships through, Annexure 2 is a scan carrying a handwritten margin note, initialled by a QA head who left in 2024, and that note raises the acceptable limit.

The lane. Priya had a feeling she had seen this before. She asked in the reviewers’ WhatsApp group and somebody remembered two similar events on the same route with the same carrier in June. Sentinel will show you one shipment at a time. It will not show you a lane, a carrier and a reefer unit together, and it certainly will not tell you how those earlier cases were closed or on what grounds.

The carrier. The consignment note said “Tapti Exp Cargo”. There is a contract with Tapti Express Cargo dated August 2021, which predates our cold chain schedule and therefore contains no temperature deviation clause at all. Whether that mattered depended on whether the carrier caused the excursion, which depended on the setpoint record, which arrives from that carrier as a CSV attached to an email, when it arrives.

Priya spent an hour and fifty minutes on this case, across seven systems and one WhatsApp thread, and released it with a note. The note is not hedging. This product's allowance is cumulative across its shelf life, so every excursion has to be written against the batch whether or not it changes the answer today.

Here is the part worth sitting with. None of those four reasons changed the outcome. The stability position she found in the first ten minutes was the answer, and it is the answer she signed. What the other hundred minutes bought was the right to sign it: establishing that the product reference really did point at this product, that the revision in force did not say something different, that the lane had no pattern behind it, and that the contract question changed who pays but not what happens to the product. That is the shape of this job. The decisions are mostly not hard. Assembling enough evidence to be allowed to make them is the whole cost.

Four weeks later the customer's QA director asked why, and it took a day and a half to reassemble the answer, because the answer had never been recorded anywhere except as the string RELEASE_WITH_NOTE in a database column.

Two thousand of those a month, and the twelve questions on tab 10 of the pack are the ones reviewers say they have to answer every time.

SECTION 3

The ten months behind this brief

Six things happened between November and August. They are recorded in full on tab 14 of the pack; this is the short version.

In **November** somebody edited a piece of wording in the recommendation assistant's prompt to suit one European customer, shipped it on a Thursday, and the mix of recommendations changed for thirty-one of our thirty-four customers. Destroy recommendations went from eleven per cent of closed cases to nineteen. We found out nine days later because a reviewer happened to query two decisions. Two of those destroy recommendations had been accepted on product that was later shown to have been inside its stability envelope the whole time. USD 310,000 of insulin went into an incinerator for no reason.

In **February** a chiller controller failed at Whitefield Cold Store 2 and emitted 1,912 excursion alerts in thirty-eight minutes. The engine auto-closed most of them, and the six hundred or so it would not left the review queue unusable for thirty-one hours. Four genuine cases belonging to other customers breached their 72-hour reporting window while sitting behind it.

In **March** a European inspector at Aurastat Biologics asked us to produce, for a disposition dated 14 August 2025, the SOP revision in force at the origin site on that date, the stability data relied upon, the temperature series used, and the identity of the approver. We produced the outcome and the approver. We could not produce the rest. A major observation was raised against the customer and attributed to us. That renewal is now conditional.

In **April** our logger vendor issued a calibration bulletin: one firmware build had been under-reporting by as much as 1.8 °C. Two hundred and fourteen cases closed in the preceding three weeks had relied on readings from affected devices. Sixty-one of them, re-adjudicated, reached a different answer. Nine had already been invoiced to carriers and those invoices had to be withdrawn.

In **June** our model provider gave us thirty days' notice on the version we were running. Re-qualifying a replacement took seven weeks, so for nearly three of them we ran on something that had not finished qualification. We told two customers. One switched automated dispositions off entirely and has not switched them back on.

In **August** a lookup built to surface similar past cases put a Meridian Bio case inside a draft justification being prepared for Pinnacle Therapeutics. They are direct competitors and we host both. A reviewer noticed an unfamiliar batch reference before the justification left the building.

SECTION 4

What people here say

From the reviewer workshop, the board paper, and conversations over the last few weeks. Quoted as said.

> *“Three of my largest customers have told me this year that our compliance evidence is the weakest thing we sell. One of them lost a hospital tender partly because of us. I am not interested in an argument about how messy our data is. Everybody’s data is messy. That is what we are paid for.”*
>
> **ROHAN BHATT · Chief Executive**

> *“A disposition is a regulated decision. A named human signs it and that human carries it. So I will take help with the looking-up, gladly, and I will take help with the arithmetic. What I will not take is a number with no argument behind it. If the reasoning is sound and the conclusion is wrong, I can work with that, because I will catch it. If the conclusion is right and the reasoning was invented, that is worse than useless to me; I cannot tell that case apart from the next one, where the conclusion is also invented and happens to be wrong.”*
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> **DR ANJALI RAO · VP Quality & Regulatory**

> *“I do not spend my day deciding. I spend it looking things up. And the thing I look up most often is what we did last time, which nobody has ever written down. Three of us keep our own spreadsheet of past cases and how we reasoned them. It is the single most useful document in this company and it lives on Deepa’s laptop.”*
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> **PRIYA MENON · Senior QA Reviewer, 9 years**

> *“Cold chain compliance is bundled into three dollars ten a consignment. It is not a line item and I am not going to make it one. The twenty-three reviewers cost one and a half million a year, and I have given the team three thousand dollars a month for the model spend, which is what is left in that line once everything else in it is paid for. I know how those two numbers look next to each other. If you think the ceiling is wrong, come and tell me with the working, because nobody has yet. And be specific about the shape of the bill, not just the size of it: it has to cost that on a quiet Sunday when nothing is going wrong and we are still paying for it.”*
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> **DANIEL OKORO · Chief Financial Officer**

> *“Two things, and neither is negotiable. Our data stays in Europe, and that is in the data processing agreement and I did not write it. And I want to know which version of whatever you are using produced a given recommendation, and to be told when that changes. In March your answer to my inspector was that you do not retain that. That is why my renewal is conditional.”*
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> **MARIEKE DE VRIES · QA Director, Aurastat Biologics (customer)**

> *“Two markets give us seventy-two hours. Brazil gives us five days. One US state programme wants it the next business day. In the last two quarters we filed fourteen late. Every single one of those was sitting in a queue behind something that turned out to be a sensor with a flat battery.”*
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> **KAVITHA REDDY · Regulatory Affairs Manager**

> *“Two thousand four hundred rules. Nobody, and I include myself, can tell you which ones fire on a given case. Every customer who asked for something different got a fork and there are eleven forks now. But that is not the part that bothers me. Last November we changed some wording and it moved the decisions for thirty-one customers. I still cannot tell you exactly why it moved. We had no way of knowing before we shipped it, and no way of knowing after.”*
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> **SANDEEP IYER · Principal Engineer, built the rules engine and the recommendation assistant, leaving in November**

> *“Clause 22 of the Meridian agreement prohibits any model, index or derived artefact built on their data from being used in the delivery of services to another customer. Pinnacle is their direct competitor and we host both. In August we came within one reviewer’s attention span of a breach with termination rights attached to it.”*
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> **NIKHIL SARIN · Commercial Counsel**

SECTION 5

What has already been tried

You should know what we have burned, so that you do not offer it back to us.

The rules engine, 2023, still in production

Two thousand four hundred rules with eleven per-tenant forks. It auto-closes about 78 per cent of the alerts that turn out to be non-actionable, which is genuinely useful. What it does not close, it escalates. Sandeep’s description above is the fair one. It is what runs today and it will keep running until something replaces it.

The recommendation assistant, 2024, still in production

On the cases the rules engine escalates rather than closes, a language model drafts the recommendation and the justification text, and the reviewer accepts it, edits it or overrules it. It sees the temperature series, the product master row and the shipment record. It does not see the SOP library, the stability repository, the contract, or anything that happened before on that lane, which is most of why reviewers overrule it 23 per cent of the time. November was a change to its prompt. June’s sunset was its model. It is what Marieke means when she asks which version produced a recommendation.

The January pilot

We tried to go the whole way: no reviewer and no rules engine, every alert straight model with a prompt, and asked for a disposition and a justification. It demonstrated extremely well and everybody in the room was impressed. Under evaluation it cited stability studies that do not exist, quoted SOP revisions that had been withdrawn two years earlier, and had no awareness whatsoever of anything that had previously happened on the lane. We stopped it in March.

The document question-answering tool, June

A vendor product pointed at the SOP library. It answers questions about a single document well. It cannot answer a question that spans two documents, and it cannot connect anything it reads to a shipment, a batch, a facility or a contract. We still pay for it. Four people use it.

Deepa’s spreadsheet

Eleven hundred rows of past cases and the reasoning behind them, maintained privately by three reviewers since 2023. It is the fastest way in this company to answer “what did we do last time”. It has no schema, no backup and one owner, and she is on maternity leave from this coming January.

SECTION 6

What the board has asked for

Board paper BP-26-09 was approved on 2 September. It is written in outcomes rather than mechanisms, on purpose. These are the numbers and conditions it commits us to.

	Commitment	Where it stands today
1	Median time from alert to signed disposition under 24 hours, by the end of the first full quarter after go-live.	6.2 days
2	At least 70 per cent of alerts closed without reaching a person at all.	About 64 per cent, via the rules engine, of a population that is 82 per cent non-actionable
3	For any disposition, the full basis on which it was reached, produced on request within the hour, for seven years after the event.	Not possible. See March.
4	Inside the cost ceiling on tab 13 of the pack.	Current adjudication cost is USD 1.51M a year in QA salary
5	No notifiable deviation notified to a regulator outside its window.	14 late in the last two quarters
6	First customer live within 90 days of design sign-off. All 34 within nine months.	n/a

Two conditions were attached to the approval, both of which the board minuted verbatim.

Dr Rao’s condition

“I will not sign a go-live until I have been shown, for a body of past cases whose outcomes we already know, what this thing would have decided, and specifically where it would have decided differently from us, and why. And I want to see that again before every change you make to it after that, not just the first time. I am willing to accept automation on the closures. I am not willing to accept it on the decisions.”

From the board minute

“We would rather ship this to one customer and be certain than to thirty and be sorry.”

SECTION 7

What you have to work with

The companion workbook is a real extract, assembled by hand over four days from nine production systems, because there is nowhere these things can be queried together. Fifteen tabs: the alert records themselves, the product and facility and carrier masters as the source systems actually hold them, the controlled document and stability indexes, the alias notes reviewers have made, reviewer activity on every case a human opened, the twenty-four cases where a reviewer wrote down why they overruled the engine, the twelve questions reviewers have to answer, the systems inventory, eight eight customers’ contractual constraints, the volume and cost figures with the model price list procurement gave us, and the incident log.

Read the README tab before the others. It says how the extract was assembled and lists what we already know is wrong with it.

The one thing worth stating plainly, because it is not obvious from any single tab: there is no shared data model across those nine systems and there never has been. A single physical facility

appears five times under five names. A product appears under a brand name, an international non-proprietary name, an internal SKU, a GTIN and two or three spellings of the strength. A carrier and its foreign subsidiary look almost identical on a consignment note and are governed by different contracts with different claim windows. The mapping between all of these lives in reviewers' heads, and partially (unverified, unmaintained and in places wrong) on tab 07.

SECTION 8

What we would like you to produce

Four things. There is no template for any of them.

A. Where you land, and why

Two to four pages. What you would build, what you would deliberately leave alone, and in what order. We would like to see your own statement of what this system has to do and what it has to be, and against each item, what in this document or the pack led you there. Where the pack is silent or contradicts itself and you had to take a position, say so and say what position you took. If you conclude that some part of what the board has asked for is the wrong thing to want, argue that.

B. The design

No page limit. Two audiences: a team of six who have to build it without coming back to you with questions, and a reviewer whose job is to find where it breaks. So both the shape of the thing (what the pieces are, what crosses the boundaries between them, what happens when a piece is unavailable), and the parts that have to be pinned down exactly: data, interfaces, state, ordering, failure and recovery. Internally we call these the HLD and the LLD; use whatever headings you prefer. Diagrams are welcome and will be read closely.

C. Something running

Pick the part of this that you think is hardest to be *right* about, and make that part real. Not the part that demonstrates best; the part that would keep you up at night. Use whatever fraction of the data pack you need and stub the rest. Show us how you know it works, and how you would know if it stopped working at three in the morning six months from now.

D. Your working log

We expect you to use AI heavily for all three of the above (for thinking the problem through, for the design, and for the build), and we would like to see how. Keep the sessions. Tell us what you kept, what you threw away, where it was confidently wrong, and how you caught it. Where you disagreed with it and were right, we want that; where you disagreed with it and were wrong, we want that more. We are at least as interested in the discarding as in the keeping, because that is the part that does not show up in the finished artefact.

SECTION 9

The conversation afterwards

Ninety minutes, with Dr Rao, Sandeep and two engineers from the platform team. The first twenty minutes are yours to walk us through where you landed. Then we choose two things from your design and go as deep as you are willing to go on them; assume we will choose the two you are least comfortable with. In the last twenty minutes we will put a change in front of you that you have not seen and ask you how your design absorbs it. There is no right answer to that part and we are not looking for one.

SECTION 10

Ground rules

• **Five calendar days from the moment this lands in your inbox.** We suggest somewhere between ten and fourteen hours inside that. We would rather read twelve hours of judgement than forty hours of output, and we can usually tell the difference.

• **There is deliberately more here than you can finish.** Deciding what not to do, and saying why, scores better than running out of time with everything half done. A short submission that draws its own boundary is a strong one.

• **Any tools, models, libraries, languages or services you like.** Nothing is off-limits and nothing is expected. If you use something unusual, tell us why.

• **The pack has real problems in it.** Parts of it contradict other parts. Parts of it are simply wrong. We have listed the ones we know about on the README tab and we have certainly not listed all of them. Noticing is part of the exercise; cleaning all of it up is not.

• **Ask anything factual.** Sudhanshu, at sudhanshu@fourkites.com, usually answers within a few hours. We will tell you what is true about Kestrel. We will not tell you what we think the answer is, and asking does not count against you.

• **None of this is a trick.** It is untidy in the specific ways it is untidy here.

• **Send your submission to the same address**, as a single archive or as a link to a repository we can read. Any format you like for the written parts; send code as files rather than pasted into a document. Tell us in the covering note how long you spent.
